

QCT Supplementary Material 11

Neutrino Physics

Content Pending, Publication-Ready Shell

Andrew Lizzio

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Abstract

Quantum Cell Theory (QCT) is a foundational framework derived from three postulates: **P1** (the universe comprises a finite number N of discrete energy cells, each carrying rest energy ε_0 , with cell count and vibrational energy conserved); **P2** (each cell resizes in proportion to local vibrational energy density, isotropically and scale-free); and **P3** (one time quantum τ_0 elapses per wave-propagation step).

QCT derives neutrino masses and mixing from the soliton sector geometry of the D_5 Cartan domain. This document will contain QCT derivations of neutrino masses, mixing angles, and oscillation parameters from P1/P2/P3. Derivations are in progress; this shell establishes the module structure.

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About This Module and the QCT Series

This document is part of the QCT Supplementary series of standalone modules. The series is:

- **QCT-Supp-0-Map-v1:** Proof Map
- **QCT-Supp-1-Postulates-v1:** Postulates P1/P2/P3, Classical Mechanics, General Relativity, Quantum Mechanics
- **QCT-Supp-2-Identification-v1:** Structural Identifications ID1/ID2/ID3, Fine Structure Constant α
- **QCT-Supp-3-MassGap-v1:** Yang–Mills Existence and Mass Gap
- **QCT-Supp-4-ElectronMass-v1:** Hopf Solitons, $N_c = 20$, Electron Mass Formula
- **QCT-Supp-5-Cosmology-v1:** Cosmological derivations C1–C11, Dark Matter and Dark Energy
- **QCT-Supp-6-Spectral-v1:** Electromagnetism, Force Mechanism, Weinberg Angle
- **QCT-Supp-7-Chemistry-v1:** Chemical and Molecular Predictions
- **QCT-Supp-8-ParticleMasses-v1:** Particle Mass Spectrum
- **QCT-Supp-9-NuclearPhysics-v1:** Nuclear Physics
- **QCT-Supp-10-Neutrinos-v1:** Neutrino Physics
- **QCT-Supp-11-QuantumGravity-v1:** Quantum Gravity and Planck Scale
- **QCT-Supp-12-DarkMatter-v1:** Dark Matter

1 Neutrino Masses from Soliton Sector Geometry

1.1 Neutrinos as Neutral Soliton Sectors of D_5

1.2 Mass Hierarchy from D_5 Stratum Structure

2 Neutrino Mixing Angles

2.1 PMNS Matrix from D_5 Symmetry Breaking

2.2 QCT Prediction of θ_{12} , θ_{23} , θ_{13}

3 Neutrino Oscillation Parameters

3.1 Mass Squared Differences from QCT

3.2 CP Violation in the Neutrino Sector

4 Majorana vs Dirac Neutrinos in QCT

4.1 Hopf Topology and the Majorana Condition